EXPLORATORY DATA ANALYSIS

CHAPTER 8

Understanding Distributions

Reading data as a whole: frequency, density, cumulative structure, shape, tails, and group comparisons

Chapter focus: frequency distributions, probability mass and density, empirical cumulative distribution functions, bins and bandwidth, multimodality, tails, and distribution comparison.

Scope: descriptive reasoning about observed distributions. Formal probability theory, inferential goodness-of-fit testing, and model fitting are outside this chapter's main objective.

Learning Objectives

- Explain what a distribution describes and why it contains more information than a single summary statistic.
- Construct and interpret frequency, relative-frequency, and cumulative-frequency distributions.
- Distinguish probability mass functions, probability density functions, and empirical distributions.
- Understand histograms, bin width, bin edges, kernel density estimates, and smoothing limitations.
- Interpret empirical cumulative distribution functions (ECDFs).
- Recognize symmetry, skewness, multimodality, gaps, heavy tails, and truncation.
- Compare distributions across groups using aligned scales and meaningful summaries.
- Use Python, Pandas, NumPy, Matplotlib's object-oriented API, and Seaborn appropriately.

Review of Earlier Chapters

Chapter 1 — What Is Exploratory Data Analysis?

EDA asks what the data contains, whether it can be trusted, and what patterns it reveals. A distribution is one of the most useful descriptions of a numerical variable because it shows how observations are spread across possible values.

Chapter 2 — Understanding the Dataset

Interpret a distribution only after identifying the observational unit, variable meaning, measurement scale, granularity, and data-generating process. A transaction-level distribution and a customer-level distribution answer different questions.

Chapter 3 — Data Quality Assessment

Invalid values, duplicates, missingness, inconsistent units, and collection limits can distort the observed distribution. A suspicious spike or gap may be a property of the process or a data-quality issue.

Chapter 4 — Exploring Numerical Variables

Mean, median, spread, quantiles, IQR, skewness, and outlier investigation summarize important aspects of numerical variables. This chapter reconnects those summaries to the complete shape of the data.

Chapter 5 — Exploring Categorical Variables

Frequency tables, counts, proportions, cardinality, and rare categories are foundational distribution tools for categorical variables. Numerical distributions use analogous ideas, but values are ordered along a numerical scale.

Chapter 6 — Exploring Date and Time Variables

Temporal event counts, durations, and inter-event gaps are variables with distributions too. Their shape can reveal bursts, long inactivity periods, and unusual timing.

Chapter 7 — Descriptive Statistics: Beyond the Basics

Quantiles, robust centers, weighted means, IQR, standard deviation, and relative variability summarize a distribution. These summaries are useful, but they cannot replace inspection of the full distribution.

Review of Chapter 7 — Descriptive Statistics: Beyond the Basics

- Choose mean and SD when they are meaningful, but compare them with median and quantiles when data are skewed or extreme.
- IQR describes the middle 50%; P05, P50, and P95 can add tail information.
- Trimmed and winsorized means are sensitivity summaries, not automatic cleaning rules.
- Weighted averages depend on the question and meaning of the weights.
- Always report valid n, missingness, units, and assumptions.
- Group summaries should include sample size and spread, not just means.

Chapter 7 Assignment and Project — Discussion Guide

No assignment answers or notebook were submitted with this request, so this section describes expected reasoning rather than grading individual work.

- For the values [4, 5, 5, 6, 7, 8, 9, 10, 12, 34], the mean is 10.1 and the median is 7.5. The high value pulls the mean upward.
- The IQR and quantiles should be calculated using the specified Pandas interpolation convention; always state that convention.
- Sample SD uses ddof=1; population SD for the observed values uses ddof=0.
- For branch averages with unequal transaction counts, the transaction-weighted mean answers the across-transaction average; the unweighted mean answers the average of branch means.
- A good toolkit should expose n, missingness, center, spread, quantiles, and explicit weighting/variance assumptions.

Expected project learning: create reusable descriptive summaries, test them on different distribution shapes, and explain why one statistic may not adequately describe a dataset.

Preview of This Chapter

We now move from a list of summary statistics to the distribution itself. The central idea is to understand how observations occupy the value range, how concentrated or dispersed they are, and how the shape changes across groups.

1. What Is a Distribution?

A distribution describes how values of a variable occur: which values are common, which are rare, how observations are spread, and where the tails lie. For a numerical variable, it connects values on a number line to their frequencies or relative frequencies.

A distribution can be described through:

- Support or range: the values the variable can take or that are observed.
- Concentration: where observations accumulate.
- Spread: how dispersed the values are.
- Shape: symmetry, skewness, peaks, valleys, and gaps.
- Tails: how observations behave at the extremes.
- Mass at particular values: repeated values, zeros, or discrete spikes.

A distribution is empirical when it is constructed from observed data. It describes the dataset in hand; it is not automatically the true population distribution.

2. Frequency Distributions

For discrete or grouped numerical values, a frequency table counts how many observations fall at each value or interval. Relative frequency divides each count by the total number of valid observations.

Score	Frequency	Relative frequency
1	2	0.10
2	3	0.15
3	7	0.35
4	5	0.25
5	3	0.15
Total	20	1.00

The denominator matters. If missing values are excluded, relative frequencies sum to one over valid observations, not necessarily over all dataset rows.

```
scores = pd.Series([1, 2, 2, 3, 3, 3, 4, 4, 5])

counts = scores.value_counts().sort_index()
relative = scores.value_counts(normalize=True).sort_index()

frequency_table = pd.DataFrame({
```

```
    "count": counts,
    "proportion": relative
})
```

For continuous measurements, exact values may be nearly unique, so grouping into intervals is often more useful. The interval boundaries and widths must be disclosed.

3. Cumulative Frequency and Relative Frequency

Cumulative frequency adds counts up to and including a value or bin. Cumulative relative frequency expresses the corresponding proportion. It answers questions such as 'What proportion of observations are at or below this threshold?'

```
counts = scores.value_counts().sort_index()
freq = counts / counts.sum()

cumulative_count = counts.cumsum()
cumulative_proportion = freq.cumsum()

table = pd.DataFrame({
    "count": counts,
    "cumulative_count": cumulative_count,
    "cumulative_proportion": cumulative_proportion
})
```

For ordered categories or continuous values, cumulative summaries are meaningful. For nominal categories such as city names, a cumulative total depends on arbitrary ordering and is generally not meaningful.

4. Histograms: Bins and Their Consequences

A histogram groups numerical observations into adjacent intervals (bins) and displays the frequency or density in each interval. Unlike a bar chart of categories, histogram bars represent numerical intervals and are conventionally adjacent.

A histogram is not a neutral picture: its appearance depends on bin width, bin edges, and range. Too few wide bins can hide structure; too many narrow bins can exaggerate random fluctuations.

```
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(values, bins=10, edgecolor="black")
ax.set_title("Histogram of Values")
ax.set_xlabel("Value")
ax.set_ylabel("Frequency")
fig.tight_layout()
```

Try several reasonable bin widths and compare the conclusions. For large datasets, automatic bin rules such as Freedman–Diaconis can provide a starting point, but domain knowledge still matters.

Freedman–Diaconis rule: bin width $h = 2 \times \text{IQR} \times n^{-1/3}$. It uses a robust measure of spread and sample size; it is a guideline, not a law.

```
q1, q3 = values.quantile([0.25, 0.75])
iqr = q3 - q1
n = values.notna().sum()

bin_width = 2 * iqr * (n ** (-1 / 3))
data_range = values.max() - values.min()
estimated_bins = int(np.ceil(data_range / bin_width))
```

Guard against zero IQR, constant data, and very small samples before applying this calculation. If IQR is zero, the rule cannot produce a useful positive width.

5. Frequency, Relative Frequency, and Density

Frequency counts observations in each bin. Relative frequency divides bin counts by the total number of observations. Density rescales heights so that the total area of histogram bars equals one.

For equal-width bins, frequency and relative frequency have similar shapes. For unequal-width bins, raw counts can mislead because wider bins naturally collect more observations. Density accounts for bin width.

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

axes[0].hist(values, bins=10, density=False, edgecolor="black")
axes[0].set_title("Frequency")

axes[1].hist(values, bins=10, density=True, edgecolor="black")
axes[1].set_title("Density")

fig.tight_layout()
```

A density height is not the probability of one exact continuous value. Probability is represented by area over an interval. For example, the area between 10 and 20 approximates the proportion of observations in that interval for a normalized empirical density histogram.

6. Probability Mass, Probability Density, and Empirical Distributions

The terminology matters:

Concept	Used for	Meaning
Probability mass function (PMF)	Discrete random variables	Probability assigned to each possible value
Probability density function (PDF)	Continuous random variables	Density; probability is area over an interval
Empirical distribution	Observed sample	Distribution formed from the observed values

A discrete variable can assign positive probability to an exact value. For a continuous variable with a density, the probability of any one exact point is zero in the mathematical model; probabilities are areas over intervals.

In EDA, a normalized histogram or KDE is an estimate/visual summary of the observed distribution. It is not proof that the population follows a particular theoretical distribution.

7. Empirical Cumulative Distribution Function (ECDF)

The empirical cumulative distribution function at x is the proportion of observed values less than or equal to x:

F■(x) = (number of observations ≤ x) / n

The ECDF is a step function. It shows the entire cumulative distribution without choosing histogram bins or a KDE bandwidth.

```
sorted_values = np.sort(values.dropna().to_numpy())
n = len(sorted_values)
ecdf_y = np.arange(1, n + 1) / n

fig, ax = plt.subplots(figsize=(8, 4))
ax.step(sorted_values, ecdf_y, where="post")
ax.set_xlabel("Value")
ax.set_ylabel("Cumulative proportion")
ax.set_title("Empirical Cumulative Distribution Function")
ax.set_ylim(0, 1.02)
ax.grid(True, alpha=0.3)
fig.tight_layout()
```

An ECDF lets you read percentiles and threshold proportions directly. If the ECDF at 50 is 0.8, roughly 80% of observations are at or below 50.

For discrete data, repeated values create vertical jumps. The exact convention at ties should be understood; the code above places the cumulative fraction after each sorted observation.

8. Kernel Density Estimation (KDE)

A KDE constructs a smooth estimate of a distribution by placing a smooth kernel around each observation and adding the contributions. The bandwidth controls smoothness.

A small bandwidth can create many narrow bumps and overstate noise. A large bandwidth can smooth away genuine clusters or peaks. KDE is a visual estimate, not a direct count.

```
import seaborn as sns

fig, ax = plt.subplots(figsize=(8, 4))
sns.kdeplot(data=values, ax=ax, bw_adjust=1)
ax.set_title("Kernel Density Estimate")
ax.set_xlabel("Value")
ax.set_ylabel("Estimated density")
fig.tight_layout()
```

KDE can behave poorly for discrete, bounded, or highly concentrated variables. It may place density outside a variable's valid range—for example, negative values for a quantity that cannot be negative. Use a histogram or ECDF alongside it, and consider whether smoothing is appropriate.

9. Reading Distribution Shape

Symmetry

A symmetric distribution has roughly mirrored shape around its center. Mean and median may be close, but equality is not a proof of symmetry.

Skewness

A right-skewed distribution has a longer or heavier right tail; a left-skewed distribution has a longer or heavier left tail. The mean is often pulled toward the long tail, but skewness should be judged from the distribution, not only from the mean–median relationship.

Unimodality and multimodality

A unimodal distribution has one dominant peak. A multimodal distribution has multiple peaks. Multiple peaks can indicate subpopulations, operational regimes, or mixtures of processes; they can also be artifacts of binning or KDE bandwidth.

Gaps and spikes

A gap may reflect a real impossible range, rounding, censoring, a collection issue, or a mixture of groups. A spike at zero may be substantively meaningful or may represent a placeholder code.

Heavy tails

Heavy-tailed data contain more extreme observations than a thin-tailed reference shape would suggest. In practice, inspect upper and lower quantiles, extremes, and the frequency of large deviations rather than relying on a single label.

10. Comparing Distributions Across Groups

Compare groups using the same measurement units, x-axis range, bin edges, and density/count convention. Separate histograms with unrelated bins can make differences appear larger or smaller than they are.

```
fig, ax = plt.subplots(figsize=(9, 4))

for name, group in df.groupby("segment"):
    ax.hist(
        group["amount"].dropna(),
        bins=20,
        alpha=0.35,
        density=True,
        label=name
    )

ax.set_title("Amount Distributions by Segment")
ax.set_xlabel("Amount")
ax.set_ylabel("Density")
ax.legend()
fig.tight_layout()
```

Overlapping histograms can become cluttered. Alternatives include ECDFs, side-by-side boxplots, violin plots, or small multiples. Use the same axes when direct comparison is the goal.

```
fig, ax = plt.subplots(figsize=(9, 4))
sns.ecdfplot(data=df, x="amount", hue="segment", ax=ax)
ax.set_title("ECDF by Segment")
fig.tight_layout()
```

An ECDF comparison is particularly useful for threshold questions: what proportion of each group lies below a specified amount? A boxplot is compact but hides multimodality and much of the distribution's shape.

11. Distribution Comparisons Need Context

- Check group sizes: a small group produces a less stable empirical distribution.
- Check missingness: groups may represent different subsets of the original population.
- Check time windows: one group may have been observed during a different period.
- Check units and measurement procedures: apparent shape differences may be measurement artifacts.
- Check whether the variable is bounded, rounded, censored, or truncated.
- Do not interpret distribution differences as causal effects without an appropriate design and evidence.

12. Discrete, Continuous, Bounded, and Zero-Inflated Variables

Not every numerical variable should be visualized or interpreted in the same way.

Variable type	Examples	EDA considerations
Discrete counts	Number of purchases, defects	Integer frequencies; many ties
Continuous measurements	Height, duration, temperature	Histogram, ECDF, quantiles
Bounded values	Percentages, ratings from 1–5	Respect valid limits; KDE may spill outside

Variable type	Examples	EDA considerations
Zero-inflated	Claims with many zeros, inactive days	Separate zero mass from positive values
Rounded/heaped	Age reported in whole years	Repeated values may reflect recording precision

For zero-inflated variables, report the proportion of zeros separately from the distribution of positive values. A single histogram can hide the distinction between whether an event occurs at all and how large it is when it occurs.

13. A Reusable Distribution Profile

The following function returns a compact empirical profile. It deliberately reports quantiles and zero proportion in addition to moments.

```
def distribution_profile(series):
    x = pd.to_numeric(series, errors="coerce")
    v = x.dropna()

    if v.empty:
        return {"rows": len(x), "valid_n": 0,
                "missing_n": int(x.isna().sum())}

    q = v.quantile([.01, .05, .25, .50, .75, .95, .99])

    return {
        "rows": len(x),
        "valid_n": len(v),
        "missing_n": int(x.isna().sum()),
        "mean": v.mean(),
        "sd": v.std(),
        "min": v.min(),
        "p01": q.loc[.01],
        "p05": q.loc[.05],
        "q1": q.loc[.25],
        "median": q.loc[.50],
        "q3": q.loc[.75],
        "p95": q.loc[.95],
        "p99": q.loc[.99],
        "max": v.max(),
        "zero_proportion": (v == 0).mean(),
        "skewness": v.skew()
    }
```

This is a descriptive starting point. For categorical variables, use frequency tables; for heavily discrete variables, include value counts. Do not assume a numeric profile alone captures the shape.

14. Common Mistakes

- Treating a histogram as if its shape were independent of bin width and bin edges.
- Confusing frequency, relative frequency, and density.
- Interpreting a KDE peak as an exact probability or count.
- Using KDE without considering discreteness, bounds, or bandwidth.
- Assuming a mean near a median proves symmetry.
- Calling every multi-peak curve a true mixture without checking binning and group composition.
- Comparing groups with different axes, bins, or observation windows.
- Ignoring missing values, zero inflation, rounding, or data collection limits.
- Assuming an empirical distribution is the exact population distribution.

Chapter Summary

- A distribution describes the full pattern of values, not just its center.
- Frequency and relative-frequency tables summarize counts and proportions.
- Cumulative distributions answer threshold and percentile questions.
- Histograms depend on bin choices; density is area-normalized, not a point probability.
- A PMF describes discrete probability mass; a PDF describes continuous density; an ECDF describes observed sample values.
- KDE smoothness depends on bandwidth and can mislead for discrete or bounded data.

- ECDFs display cumulative structure without bin choices.
- Shape includes symmetry, skewness, modality, gaps, spikes, and tails.
- Group comparisons require consistent scales, bins, windows, and context.

Assignment

Part A — Conceptual Questions

- Explain the difference between frequency, relative frequency, cumulative frequency, and density.
- Why does histogram bin width affect interpretation?
- What does the area under a normalized density histogram represent?
- Distinguish a PMF, PDF, and ECDF.
- Why is the probability density at x not the same as the probability of observing exactly x in a continuous model?
- What does an ECDF show that a histogram may obscure?
- Explain the effect of KDE bandwidth that is too small or too large.
- Give three reasons a multi-modal appearance might arise.
- Why should group distributions use common bins and axes?
- How should you investigate a variable with a large proportion of zeros?

Part B — Practical Analysis

Use the following values:

```
x = [0, 0, 1, 1, 2, 2, 2, 3, 4, 5, 8, 13, 21, 34]
```

- Construct a frequency and relative-frequency table.
- Construct a cumulative relative-frequency table.
- Plot a histogram using at least three reasonable bin choices. Explain what changes and what remains stable.
- Plot an ECDF and use it to estimate the proportion of values at or below 5.
- Describe the distribution's concentration, gaps, and tail behavior.
- Explain why a single mean and SD would not fully describe this dataset.

Part C — Group Comparison

Use a real dataset with a numerical variable and a categorical grouping variable.

- Create group-wise histograms with shared bin edges and density scaling.
- Create an ECDF comparison.
- Compare group sizes, medians, IQRs, tails, and zero proportions where relevant.
- Identify at least two similarities and two differences.
- List at least three contextual or data-quality explanations that should be investigated before interpreting the differences.

Part D — Coding Assignment

- Implement a distribution_profile() function with n, missingness, quantiles, mean, SD, min/max, skewness, and zero proportion.
- Generate frequency, relative-frequency, and cumulative-frequency summaries where appropriate.
- Create histogram, ECDF, and KDE views; explain the limitations of each.
- Compare at least two groups using consistent scales and bins.
- Test the workflow on discrete, continuous, bounded, and zero-inflated variables.

Coding Project — Distribution Exploration Module

Extend the cumulative EDA master notebook with a distribution-exploration module. Integrate it with the numerical summary toolkit from Chapter 7.

- Create reusable functions for frequency tables, cumulative distributions, and distribution profiles.
- Build a histogram helper that accepts explicit bins or a documented bin-selection rule.
- Build an ECDF helper that handles missing values and ties correctly.
- Add optional KDE visualization with documented bandwidth and warnings for discrete/bounded data.
- Build group-comparison plots using common bin edges, scales, and clearly stated normalization.
- Test on at least four data shapes: approximately symmetric, skewed, multimodal, and zero-inflated or bounded.
- Write interpretations that distinguish observed patterns from hypotheses about their causes.
- Record sample size, missingness, units, and visualization parameters for reproducibility.

Expected reusable components: distribution_profile(), frequency_table(), cumulative_distribution(), plot_histogram(), plot_ecdf(), and a cautious KDE helper. Keep plotting functions compatible with Matplotlib's object-oriented API and pass Axes objects when practical.

Deliverables: updated cumulative EDA master notebook, a short distribution-comparison report, and exported plots.

Preview of Next Chapter

Chapter 9 — Distribution Shape: Skewness, Kurtosis, and Quantiles

We'll examine distribution shape more rigorously: moment-based and quantile-based skewness, kurtosis and tail weight, robust shape measures, and the limits of interpreting these statistics. We'll connect numerical shape measures to histograms, ECDFs, and real-world data behavior.